Song Fei

PhD Student @ HKUST(GZ) · AIGC & Low Level Vision

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Research Summary

PhD student specializing in Generative AI and Low Level Vision. I build systems that define the State-of-the-Art:

- SOTA Visual Frontiers: Pioneered LucidFlux and UltraFlux, pushing visual generation quality beyond strong commercial baselines.

Education

The Hong Kong University of Science and Technology, Guangzhou — Guangzhou, China
PhD Student — Aug. 2026 – Present

- Advisor: Prof. Lei Zhu
- Research Focus: Text-to-Image Generation, Visual Restoration.

The Hong Kong University of Science and Technology, Guangzhou — Guangzhou, China
MPhil in Robotics and Autonomous Systems (ROAS) — Aug. 2024 – Aug. 2026

- Advisor: Prof. Lei Zhu and Prof. Jun Ma
- Research Focus: Text-to-Image Generation, Visual Restoration.

Northeastern University — Shenyang, China
B.Eng. in Industrial Intelligence — Sep. 2020 – Jul. 2024

Research Experience

JD Retail — Beijing, China
TGT (Tech Genius Team) Research Intern — Jun. 2026 – Present

- E-commerce Video Generation: Worked on post-training for multi-reference image-conditioned and audio-driven video generation, improving product fidelity, temporal consistency, and audio-visual synchronization for retail content creation.

OPPO Research Institute — Shenzhen, China
Research Scientist Intern (Image Super-Resolution & Restoration) — Dec. 2025 – Present

- Image Restoration & Super-Resolution: Contributed to the research and development of advanced generative models for image super-resolution and degradation restoration, targeting real-world mobile photography scenarios. The proposed methods improved perceptual quality and detail reconstruction under low-light and compressed conditions.

The Hong Kong University of Science and Technology, Guangzhou — Guangzhou, China
MPhil Student — Aug. 2024 – Aug. 2026

- Visual Frontiers: Served as a core contributor to LucidFlux (ICLR'26) and UltraFlux (CVPR'26). LucidFlux pushes the boundary of visual quality, surpassing strong commercial baselines (e.g., Meitu SR), and has been successfully launched as an Official API on Fal.AI.

Selected Publications

Summary: Published 8+ papers in top-tier conferences/journals (CVPR, ICCV, ICLR, NeurIPS, MICCAI), including 4 first-author/co-first-author papers.

* Equal Contribution.

First-Author & Co-First-Author Publications

[CVPR 2026 Highlight] UltraFlux: Data-Model Co-Design for High-quality Native 4K Text-to-Image Generation across Diverse Aspect Ratios
Tian Ye*, Song Fei*, Lei Zhu.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[ICLR 2026] LucidFlux: Caption-Free Photo-Realistic Image Restoration via a Large-Scale Diffusion Transformer
Song Fei*, Tian Ye*, Lujia Wang, Lei Zhu.
International Conference on Learning Representations (ICLR), 2026.

[MICCAI 2026] S2DLight: Single Step Diffusion Network via Degradation-Aware Adaptation for Surgical Endoscopic Image Low-Light Enhancement
Song Fei, Guanyi Wang, Tian Ye, Sixiang Chen, Jianyu Lai, Lei Zhu.
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.

[NeurIPS 2026] LucidNFT: LR-Anchored Multi-Reward Preference Optimization for Flow-Based Real-World Super-Resolution
Song Fei*, Tian Ye*, Sixiang Chen, Zhaohu Xing, Jianyu Lai, Lei Zhu.
Neural Information Processing Systems (NeurIPS), 2026.

Co-Authored Publications

[MICCAI 2026] MoSE: Mixture-of-Scale-Experts for Medical Image Restoration
Xingyu Li, Zhaohu Xing, Zekai Liu, Shuaibo Li, Hongqiu Wang, Song Fei, Lei Zhu.
International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.

[ICLR 2026] PosterCraft: Rethinking High-Quality Aesthetic Poster Generation in a Unified Framework
SiXiang Chen*, Jianyu Lai*, Jialin Gao*, Tian Ye, Haoyu Chen, Hengyu Shi, Shitong Shao, Yunlong Lin, Song Fei, Zhaohu Xing, Yeying Jin, Junfeng Luo, Xiaoming Wei, Lei Zhu.
International Conference on Learning Representations (ICLR), 2026.

[CVPR 2025] SnowMaster: Comprehensive Real-world Image Desnowing via MLLM
Jianyu Lai, Sixiang Chen, Yunlong Lin, Tian Ye, Yun Liu, Song Fei, Zhaohu Xing, Hongtao Wu, Weiming Wang, Lei Zhu.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

[ICCV 2025] GenHaze: Pioneering Controllable One-Step Realistic Haze Generation for Real-World Dehazing
Sixiang Chen, Tian Ye, Yunlong Lin, Yeying Jin, Yijun Yang, Haoyu Chen, Jianyu Lai, Song Fei, Zhaohu Xing, Fugee Tsung, Lei Zhu.
International Conference on Computer Vision (ICCV), 2025.

Honors & Academic Service

Honors & Awards

- Postgraduate Scholarship of HKUST(GZ), 2024–2026

Academic Service

- Conference Reviewer: AAAI 2026/2027, BMVC 2026, NeurIPS 2026, ICLR 2027.